

# Chen-An Wang

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## Experience

**07/2024-08/2024 Fortune Medical Instrument Corp.**

**Title: Automation Engineer**

**Project: Automation of silicone component fabrication**

My role was to specify and order equipment, coordinate with inhouse maintenance/operations, supervise (or self perform) installation, connecting wiring and piping for all components, programming PLCs and SCADA for optimal quality, develop written maintenance and safety procedures, and provide training for a new automated production line. The system upgrade quadrupled the throughput rate and eliminated the existing assembly line bottleneck. Additional tasks performed during the same internship included maintenance of legacy equipment and creation of wiring diagrams and operation manuals to replace lost documentation. Work was performed in a clean room environment with strict adherence to all cleanroom protocols.

**Project duration on site 160 hours**

**07/2023-08/2023 Fortune Medical Instrument Corp.**

**Title: Management Information System Engineer**

**Project: Development of an internal website connecting sales to a master database**

Collaborated with the MIS department while creating an internal website for the sales department. I built the product using HTML and CSS for developing the front end and C# for the backend, this allowed sales to connect their data directly to an existing company database.

After learning the details of my upcoming internship, I refreshed my coding skills prior to arrival on site. Due to the short duration of the internship and the significant amount of effort required my review was able to make a significant impact. I was not only able to complete the project on time, but I was able to perform a thorough beta testing of all known edge cases. Complete code commenting and documentation were delivered prior to my departure.

**Project Duration: on site 320 hours, preparation before arrival 150 hours**

**07/2022-08/2022 Fortune Medical Instrument Corp.**

**Title: Production Team Member**

**Project: Development and testing of a Negative Pressure Wound Therapy (NPWT) system**

Collaborated with design engineers and company principles in the development of a new product that utilizes vacuum pumping, liquid collection, and precision control systems. The system extracts purulence, while keeping the wound area sterile and open, this helps to accelerate healing. My responsibilities included mechanical and electrical design input, wiring harness design, circuit testing, and reliability analysis. I also developed software both for the device, and the testing fixtures that confirm proper operation of the user interface. After completion of initial testing, the team and I hosted a product launch event. The event successfully opened opportunities with new customers while also fostering a deep sense of collaboration with existing customers.

**Project Duration: on site 240 hours, preparation before arrival 85 hours**

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## Education

09/2024-06/2028 University of California San Diego (UCSD) Data Science, San Diego, United States  
09/2020-06/2024 Kang Chiao International School (KCIS), New Taipei City, Taiwan

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## Activities

### **Computer Science Training**

#### **07/2023 National Yang Ming Chiao Tung University Computer Science / AI Camp**

Completed a NYCU Computer Science Camp, where Python coding was extensively used on coding a basic Reversi game and its implemented computational AI. Tuned user interface to implement online player vs player, player vs AI, and AI vs AI game mechanics. Gained practical experience in game logic, algorithm design, and debugging.

Total: 40 hours

#### **09/2022-01/2023 Kang Chiao Python Course & Certificate of Introduction to Programming Using Python by Microsoft Technology Association**

Continued on the Python course and earned a certificate in Python programming. Explored basics of image recognition services like OpenCV and bridging such skills to hardware components like cameras and sensors. The exam contained general scripting and data handling procedures. Mainly prepared me for the college course and is a valuable skill set.

Total: 30 hours

#### **09/2022-01/2023 Kang Chiao AutoCAD Course**

Learned core AutoCAD skills for 2D and 3D drafting and engineering drawings. Filling up missing knowledge of hardware designs and getting hands-on experiences with laser cutting and 3D printing. Practiced mechanical sketches and produced products for daily use.

Total: 20 hours

#### **08/2021 Chang Gung University Computer Science Camp Online**

Introduction to transformer algorithms and language models. Course work includes practices of partial AI codes, project structures, and commonly used techniques supporting immense databases of knowledge. Explored possible career paths and relevant data-driven projects.

Total: 15 hours

#### **07/05/2021-07/15/2021 CodingBar Python Course**

Completed an intensive course on basic Python programming in high school, aiming for obtaining coding certification. Learned general insights on basic data structures, problem solving tactics, and project management.

Total: 15 hours

## **Community Engagement**

### **08/2020 Flip School**

"Pley School" is an organization that fosters learning through hands-on activities for children. Performed field research to local, small businesses to gain insights of management on traditional industry. Identified challenges they faced and interviewed with the owner to propose viable solutions and culminating in presentations.

Total: 40 hours

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## **Volunteer Work**

### **07/2021 Beach Cleanup, KCIS**

Participated in a community beach cleanup, collected and disposed of litter while following safety and waste-handling procedures. Ran campaign and raised awareness on local waste management. Encouraged the community for continued stewardship. Engaged in courses and educated peers about the importance of environmental well-being.

Total: 15 hours